

3rd Week Summary (09/17/26)

- Probability Rules

The probability $P(A)$ of any event A satisfies $0 < P(A) < 1$.

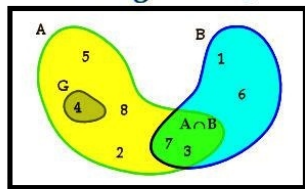
If S is the sample space, then $P(S) = 1$.

Two events A and B are **disjoint** (mutually exclusive) if they have no outcomes in common and so can never occur together. If A and B are disjoint, $P(A \text{ or } B) = P(A \cup B) = P(A) + P(B)$

For any event A , $P(A \text{ does not occur}) = P(\bar{A}) = 1 - P(A)$.

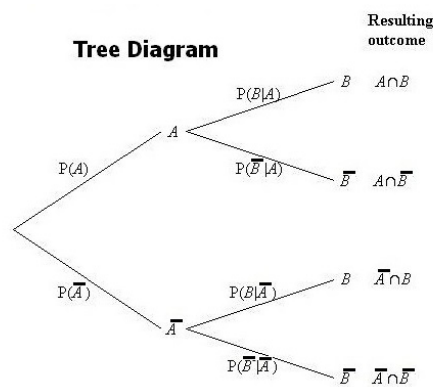
- Addition rule in general : $P(A \text{ or } B) = P(A \cup B) = P(A) + P(B) - P(A \cap B)$

Venn Diagram set operations



$A = \{5, 8, 2, 4, 7, 3\}$
 $B = \{1, 6, 7, 3\}$
 $G = \{4\}$
 $A \cap B = \{7, 3\}$
 $A \cup B = \{1, 2, 3, 4, 5, 6, 7, 8\}$
 $G \cap B = \emptyset$

Tree Diagram



- $P(A|B)$, the **conditional probability** of A given that B has occurred, can be thought of an adjusted version of the probability of A in light of the additional information that B has occurred.

When $P(B) > 0$, the conditional probability of A given B is: $P(A|B) = \frac{P(A \text{ and } B)}{P(B)} = \frac{P(A \cap B)}{P(B)}$

- General **multiplication** rule : $P(A \text{ and } B) = P(A \cap B) = P(A|B)P(B) = P(B|A)P(A)$.
- Two events A and B that both have positive probability are **independent** if: $P(A|B) = P(A)$ or $P(A \cap B) = P(A)P(B)$.

- Bayes' Rule : $P(A|B) = \frac{P(B|A)P(A)}{P(B)} = \frac{P(B|A)P(A)}{P(B|A)P(A) + P(B|\bar{A})P(\bar{A})}$

- Law of total probability: If $A_1, A_2, \dots, A_k$ are disjoint events whose probabilities are not 0 and add to exactly 1, then:

$$P(B) = P(B|A_1)P(A_1) + \dots + P(B|A_k)P(A_k)$$

- General Bayes' Rule : Suppose that $A_1, A_2, \dots, A_k$ are disjoint events whose probabilities are not 0 and add to exactly 1, then :

$$P(A_i|B) = \frac{P(B|A_i)P(A_i)}{P(B|A_1)P(A_1) + \dots + P(B|A_k)P(A_k)}$$

- A **random variable** (*discrete* or *continuous*) is a variable whose value is a numerical outcome of a random phenomenon.

A probability model with a sample space made up of a list of individual outcomes is called **discrete**.

Binomial(n, π): $P(X = k) = \frac{n!}{k!(n-k)!} \pi^k (1-\pi)^{n-k}$, $k = 0, 1, 2, \dots, n$

Poisson(μ): $P(Y = y) = \frac{\mu^y e^{-\mu}}{y!}$, $y = 1, 2, \dots$